

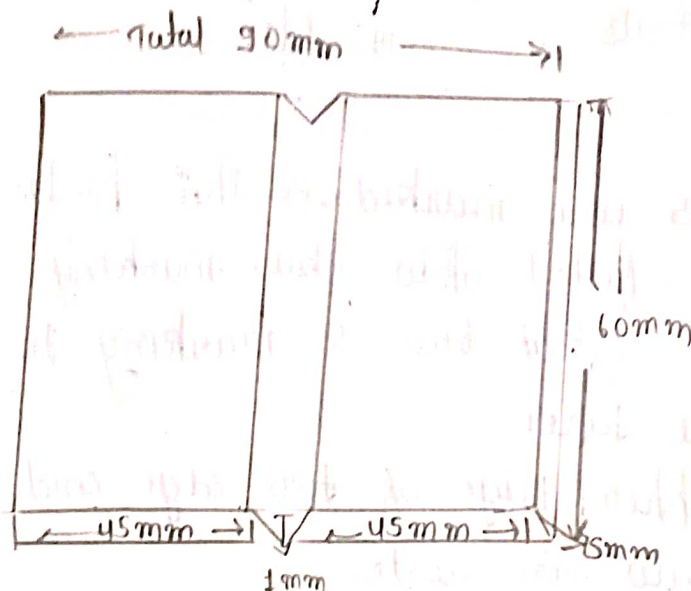
SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

Name : Chetan Salanki / WELDING - SHOP
Roll No. : 15080

TITLE : To prepare job as per drawing.

OBJECTIVES :
: To familiarise with the welding equipment.
: To aware the metal ARC welding and Gas welding.
: To study different parameter and aspect of welding.

DRAWING:



Raw material size : 90X60X5(mm) All dia in mm.

material → mild steel plate

Job name → Single V-butt joint

Machine/Tool / Hand Tool	Particular	Specification
① Electric arc welding equipment	① Electrode ① Electrode holder	
② Hand tools cutting tools	Butt-rod file triangular	iron body high carbon steel
③ measuring & marking tools	Chipping ham. (i) try square (ii) calliper (iii) file	Steel plate mild plate

Sequence of operations :

- ① Dimensions are marked on the plate as given
- ② Edges are fitted upto their marking
- ③ The plates fitted b/w 2 marking to get an angle joint
- ④ Join upper edge of two edge and weld them
- ⑤ Cool plates in water
- ⑥ Chisel away the slag with chipping hammer to finish the welding

Precautions :

- ① Use correct voltage & current
- ② All connections should be tight & insulated
- ③ The electrode should be handled & moved properly
- ④ welding space should have proper welding orientation

WELDING TORCHES

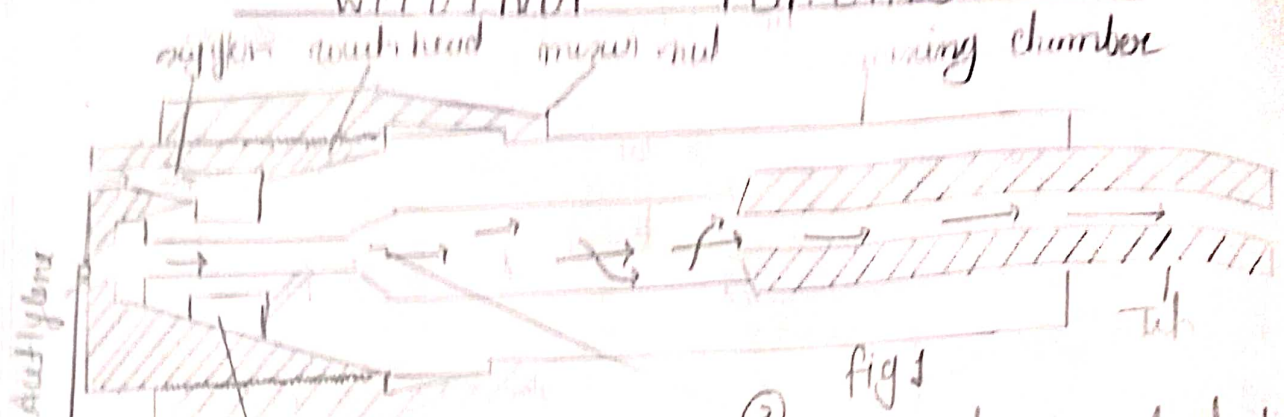


fig 1

- ① fuel gas enters at ① oxygen enters through no. of ports
 - ② around the acetylene part ③
 - ④ flow pattern in pressure forward type
- oxy - Acetylene welding torch

oxygen flow through passage ① into the venturi ② and around the fuel gas into the mixing chamber from a no. of ports

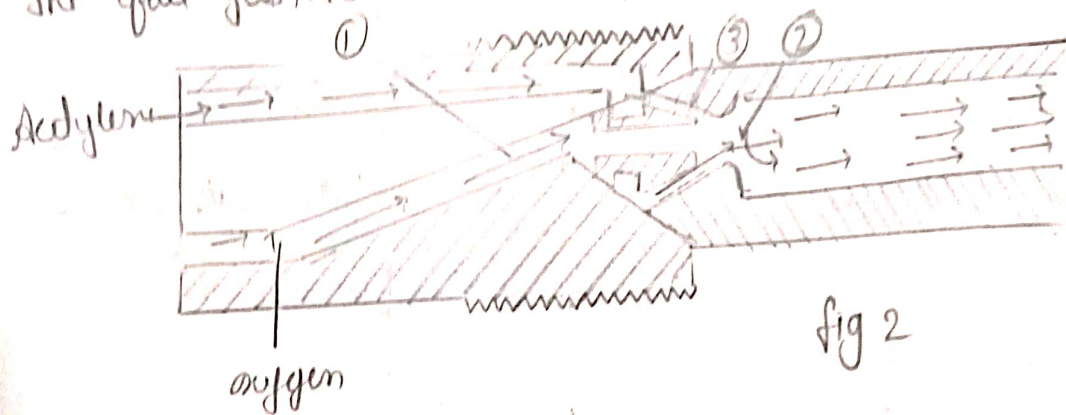


fig 2

low flow system in injector type oxy - Acetylene welding torch

oxy - Acetylene torch

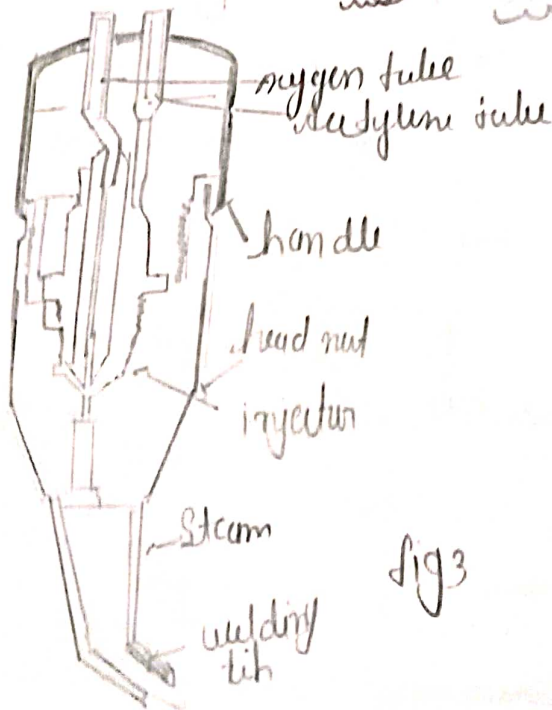
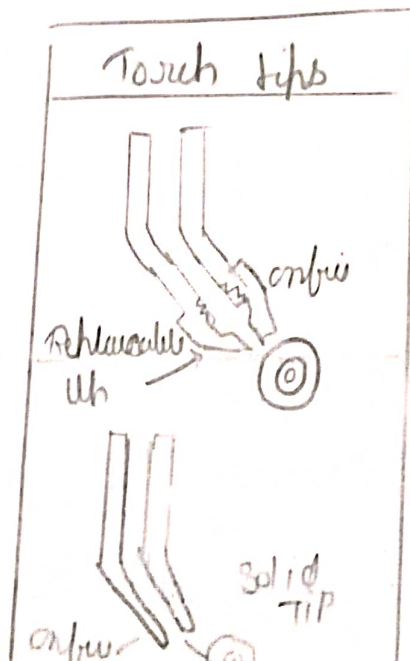
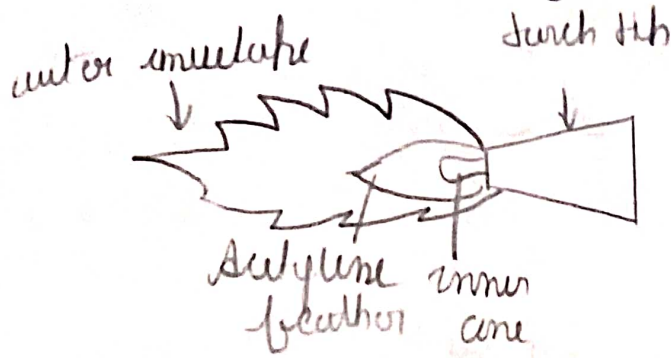


fig 3



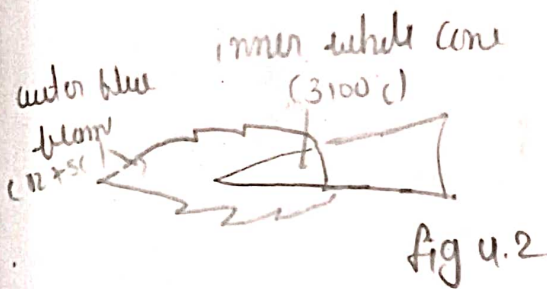
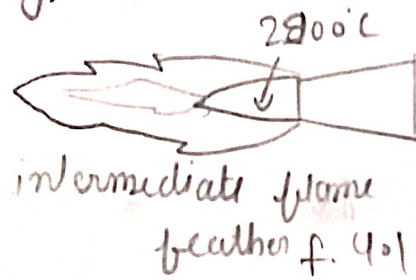
Structure of Arc - Autogenous flame



Flame type:-

A. The carburizing or Reducing flame

- Excess acetylene
- Smoke & soot
- Suitable for welding high carbon steel and cast iron



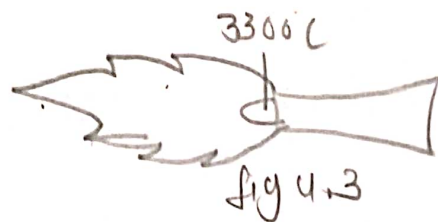
B The Neutral flame

- Equal quantities of oxygen & acetylene
- Very little smoke but hissing
- Suitable for welding of low carbon steels

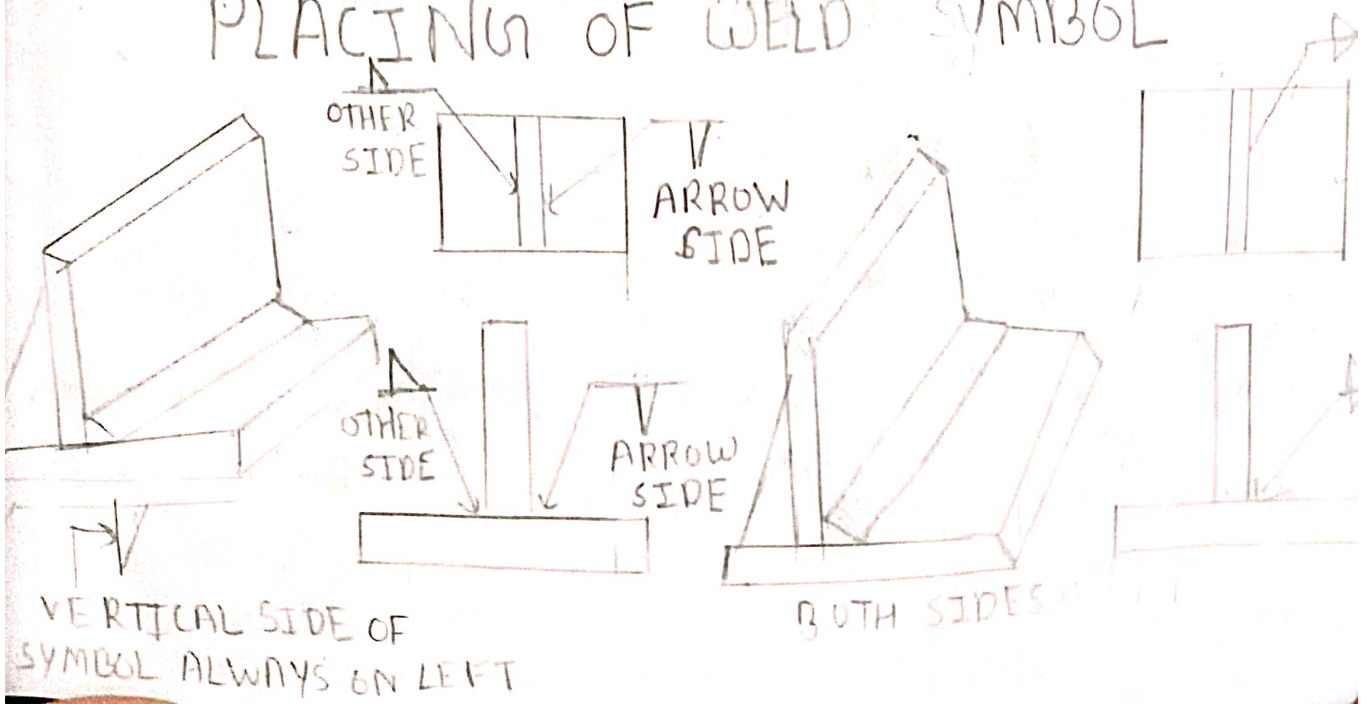
Structure starts c. 2 cu. ft. etc

C. oxidizing flame

- Excess oxygen - noisy with a blue flame
- Sooty - suitable for welding copper base alloys zinc base alloy and boron metal



PLACING OF WELD SYMBOL





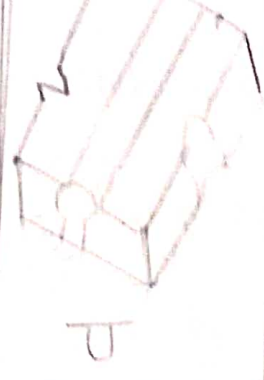
T SYMBOL
SQUARE BUTT



V
SINGLE-V
BUTT



Σ
DOUBLE-V
BUTT



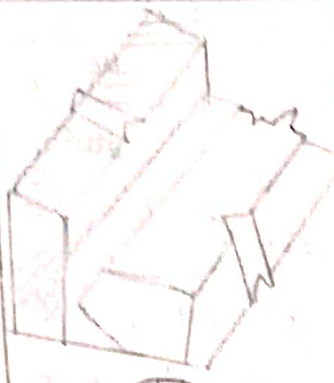
U
DOUBLE-U
BUTT



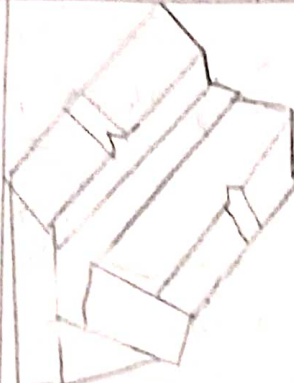
E
DOUBLE
BUTT



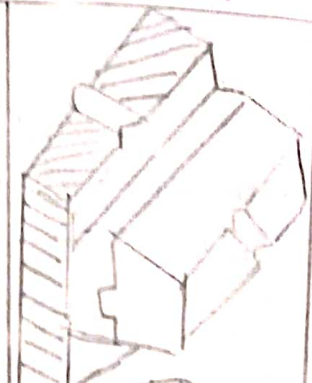
T
SINGLE-
BEVEL BUTT



E
DOUBLE-
BEVEL BUTT



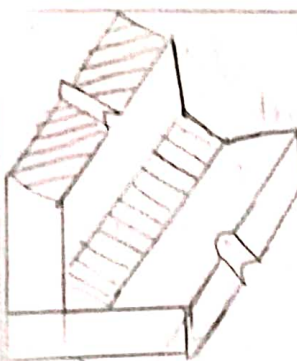
P
DOUBLE-
BEVEL BUTT



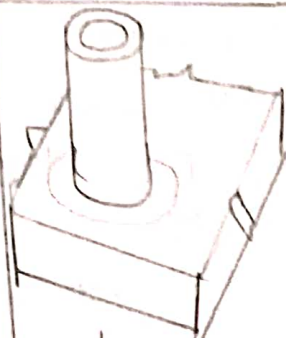
E
DOUBLE-J
BUTT



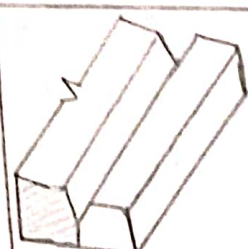
D
BEAD
OR



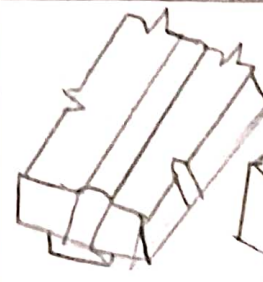
△
FILLET



T
STUD



○
SEALANT
RUN



E
BACKING STRIP

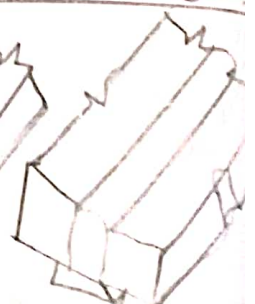
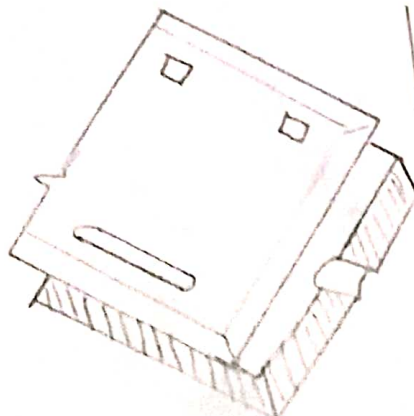
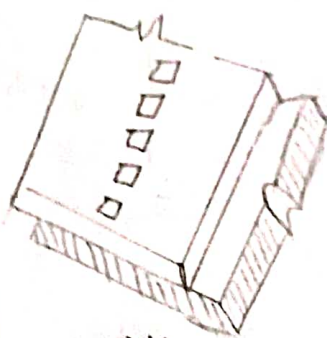


fig 5

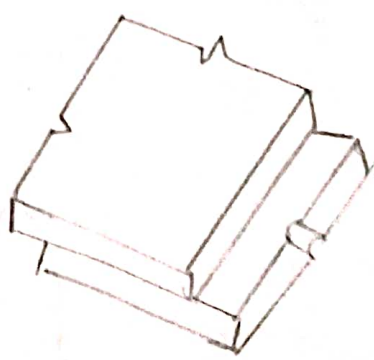
WELD SYMBOLS - II



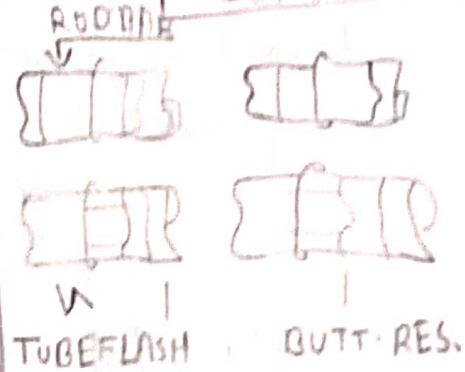
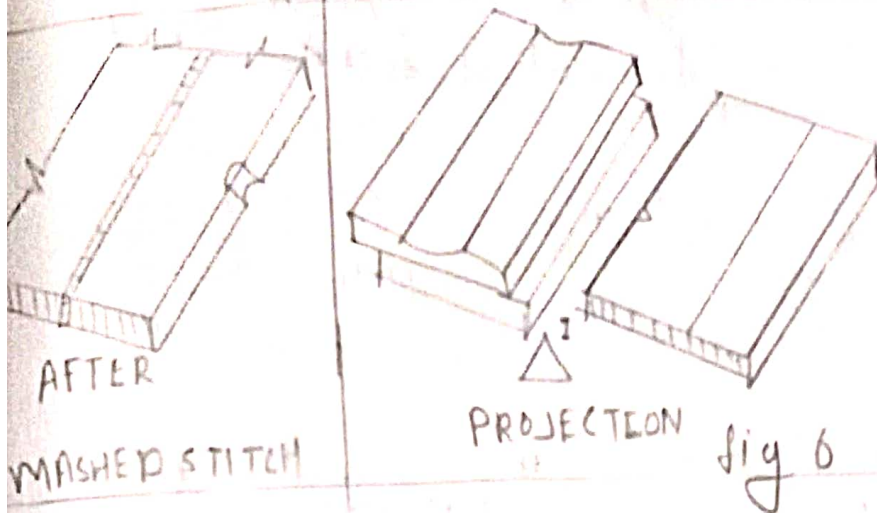
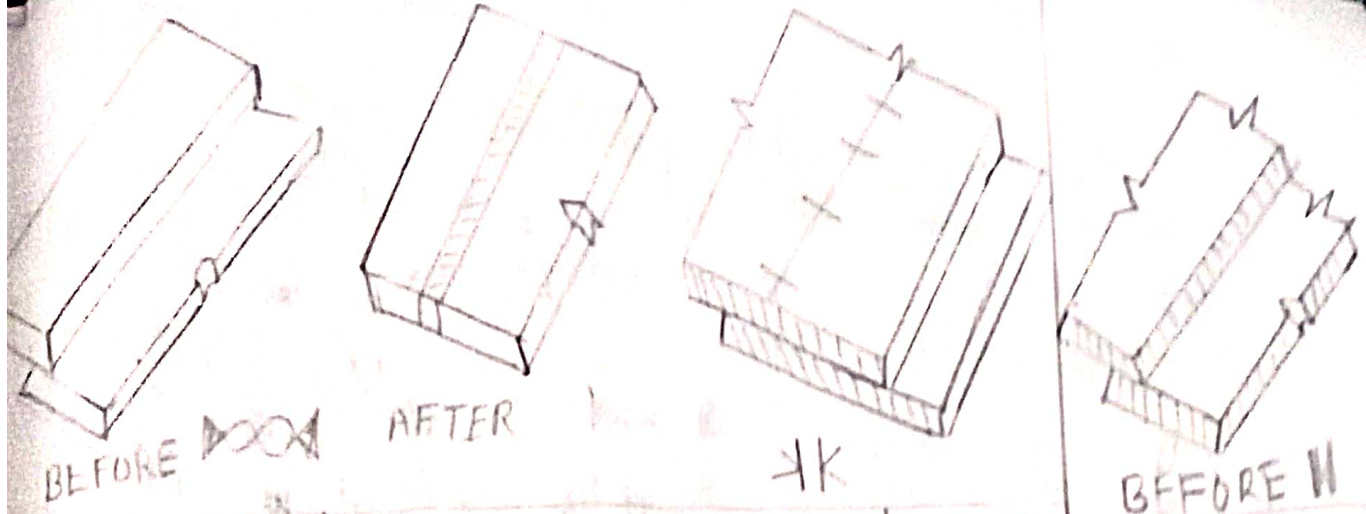
□ SYMBOL
PLUG



X
SPOT



X
SEAM



hack Saw



fig 7

Tong



fig 8

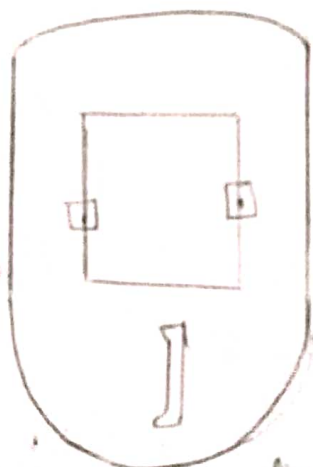


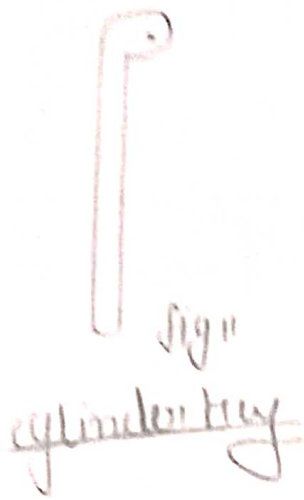
fig 9

welding screen



fig 10

Scraper



Auger bit
pipe reel

oddie
caliper

Divider



welding goggles



Try Square



Electric arc holder



Power Cable
600 AMP



Hand Glove



Ans 1 → It is a type of welding that uses a welding power supply to create an electric arc between an electrode & the base material to melt the metals at the welding point. They can use either AC or DC current & consumable or non consumable electrodes.

Ans 2 → (i) Used in fabrication works

(ii) used in construction while joining joints

(iii) used in boilers

(iv) used in air & water tank

(v) plasma arc welding is used for cutting electric machine

Ans 4 → The heat of the arc melts the flux which generates a gaseous shield to push air away from the molten pool & also form ingredients used with various impurities such as surface oxides creating slag which floats on the surface of the weld pool forming a crust which protects the weld while it is cooling.

Ans 5 → A welding arc is started when the ^{welding} current is forced across a gap b/w the electrode tip and the base metal. A welder must be able to strike and establish the correct arc easily and quickly.

Ans 6 → The size of electrode depends on the length of arc.

Ans 7 → The principle of gas welding is that the flame is produced by combustion gases with the help of oxygen. The pressure of the flame is to heat the metal plate to the parent metal.

Ans 8 → (a) The carburizing or reducing flame obtained by burning excess of acetylene with less amount of O_2 .

(b) The Neutral flame → obtained by burning equal proportion of oxygen and acetylene.

(c) Oxidizing flame → obtained by burning acetylene in excess of oxygen.

Ans 9 - (a) Gas welding $\rightarrow 1700^{\circ}\text{C}$ to 3200°C

(b) Arc welding $\rightarrow 6000^{\circ}\text{C}$ to 7000°C

Ans 10 $\rightarrow$ Lack of fusion

In welds in completely fused state, called lack of fusion present. A weld can lack of union with the parent metal or with a previous weld bead.

(b) Under cut

Undercutting is a groove or cracks that occurs near the toe of the weld when the weld flow causes the weld metal fails to fill that grooved area, resulting in a weak weld that is prone to cracking along.

(c) Porosity

Porosity is the weld metal contamination in the form of a trapped gas.

(d) Slag inclusion in welding

Small particles of weld flux

trapped in the weld metal which
prevents complete penetration of
the weld metal of low ductility

Ans 3 → (a) false, because arc length is
the distance through the web
center of the arc from tip of
arc the electrode to the bottom
of arc

(b) True

(c) True

(d) True